

OpenAPI Mock Server Generator — Epics & User Stories

Lab 2 — Jira Agile Backlog | Pranav Kalla, PES1UG24AM197

Epic OMSG-1: Specification Ingestion & Validation (Priority: High)

Enable the API Developer to import, parse, and validate OpenAPI 3.0 specs before mock generation.

ID	User Story	Points	Priority
OMSG-6	As an API Developer, I want to upload an OpenAPI 3.0 spec file (YAML/JSON) or import via URL, so that the system can generate mocks from it.	5	High
OMSG-7	As an API Developer, I want the system to validate the spec's structural conformance to OpenAPI 3.0, so that I know it's parseable before proceeding.	8	High
OMSG-8	As an API Developer, I want descriptive error messages with the location of syntax/schema errors, so that I can quickly fix a malformed spec.	3	Medium

Epic OMSG-2: Mock Endpoint Generation (Priority: High)

Automatically generate live, schema-accurate mock REST endpoints from the parsed spec.

ID	User Story	Points	Priority
OMSG-9	As an API Developer, I want the system to auto-create mock endpoints for every path/method in my spec, so that I don't configure each one manually.	8	High
OMSG-10	As a QA Engineer, I want mock endpoints to return randomized JSON responses matching the declared schema, so that I can test against realistic data.	8	High
OMSG-11	As a QA Engineer, I want undefined paths to return 404, so that I can verify my client handles unknown endpoints correctly.	2	Medium

Epic OMSG-3: Request Validation (Priority: High)

Validate incoming requests against schema and surface errors.

ID	User Story	Points	Priority
OMSG-12	As a QA Engineer, I want incoming requests validated against the declared schema (params, headers, body), so that I catch contract violations early.	8	High
OMSG-13	As a QA Engineer, I want a 400 response with detailed validation errors on an invalid request, so that I know exactly what's wrong.	3	High

ID	User Story	Points	Priority
OMSG-14	As a QA Engineer, I want to view a log of requests/responses (including rejected ones), so that I can review my test session history.	5	Medium

Epic OMSG-4: Response Configuration (Priority: Medium)

Let developers customize mock behavior — latency and overrides — to emulate real-world conditions.

ID	User Story	Points	Priority
OMSG-15	As an API Developer, I want to set fixed or randomized latency globally or per-endpoint, so that I can simulate real network conditions.	5	Medium
OMSG-16	As an API Developer, I want to override an endpoint's response with a custom payload, so that I can simulate specific business scenarios.	5	Medium
OMSG-17	As an API Developer, I want to force a specific status code (e.g., 401, 429, 500) on an endpoint, so that I can test my client's error handling.	2	Medium

Epic OMSG-5: Performance & Reliability (Priority: High)

Ensure the mock server is fast, stable, and secure under load.

ID	User Story	Points	Priority
OMSG-18	As an API Developer, I want the mock server to sustain $\geq 1,000$ req/s, so that it doesn't bottleneck CI pipelines.	13	High
OMSG-19	As a QA Engineer, I want configured latency to stay accurate within ± 10 ms under peak load, so that my results are trustworthy.	8	High
OMSG-20	As an API Developer, I want malformed or oversized (>5MB) spec uploads rejected gracefully, so that the server never crashes.	5	High
OMSG-21	As an API Developer, I want each mock instance isolated with sanitized inputs, so that no injection or data leak occurs.	8	High

Total: 5 Epics, 16 User Stories, 84 story points.